CSCI250 Computer Organization

Units of Time, Storage, and Throughput

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1. Introduction
   - Lab Instructions
2. Time
   - Frequency
3. Capacity
   - Memory and Storage
4. Bandwidth
   - Throughput

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Instructions for Lab Answers

- Open the provided document for Lec03.txt in Notepad or Notepad++ or the equivalent for a Mac.
- I want a .txt document not anything else!!!
- I will show you how to work a problem and then ask you to work several.
- Answer each question as you get to them in the slides - begin your answer on the line beneath the question and use tabs to indent each line of the answer.

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KHz, MHz, GHz, ... My head hurtz!

Hertz is a measure of the frequncy of a periodic event.
Named for German physicist Heinrich Rudolf Hertz.
..
One hertz means one time per second. ..
Example:
The average human heart beats 72 times per minute.
What is the frequency of the heart in hertz?
..
72 beats per minute / 60 seconds per minute =
1.2 Hertz

Cycle Time

- A hummingbird can flap its wings up to 80 times per second.
- The hummingbird flaps it's wings at 80 Hertz.
- How many seconds does it take for each flap?
  - $1/80$ times/second $= 1/80$ seconds $= 0.0125$ seconds
- The duration of time to complete one event is called the cycle time.

Cycle Time and Frequency

- A chihuahua can do one wag of its tail in 0.1 seconds.
  - 0.1 seconds / wag
- How many times does the dog wag its tail in one second?
  - 1 / (0.1 seconds / wag) = 10 wags / second
- What is the frequency of this dog's tail wagging?
  - 10 Hz
- What is the relationship between cycle time and frequency?
  - They are reciprocals.

Units of Frequency

- 1 Cycle per second = 1 Hertz = 1Hz
  - One operation per second
- 1,000 (10^3) Cycles per second = 1 Kilohertz = 1 KHz
  - 1,000 operations per second
- 1,000,000 (10^6) Cycles per second = 1 Megahertz = 1 MHz
- 1,000,000,000 (10^9) Cycles per second = 1 Gigahertz = 1 GHz
- 1 x 10^{12} Cycles per second = 1 Terahertz

Units of Time

- Second
- Millisecond = ms = 1/1000 of a second = 1 x (10^{-3}) seconds
- Microsecond = μs = 1 x (10^{-6}) seconds
- Nanosecond = ns = 1 x (10^{-9})

Units of Time Lab Exercises

You do not need a calculator. You are simply moving the decimal point. Do not skip steps, show your work.

Q1 Convert 2.5 seconds to milliseconds

Q2 Convert 0.00004 seconds to milliseconds

Q3 Convert 0.92 ms to nanoseconds

Q4 Convert 5 μs to seconds

Clock Cycle Lab Exercises

You do not need a calculator. You are simply moving the decimal point, or changing the power of 10. Do not skip steps, show your work.

Q5 A processor runs at 5 KHz, what is its cycle time?

Q6 A processor runs at 100 MHz, what is its cycle time in ns?

Q7 What is the clock speed in Hz of a processor whose cycle time is 2 x (10^{-3}) seconds?

Q8 What is the clock speed in MHz of a processor whose cycle time is 2 x (10^{-6}) seconds?

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Units of Memory or Storage

The approximate amounts are generally used rather than the exact amounts.

- 1 bit is a single binary digit
- 1 byte = 8 bits
  - Represent 1 ASCII character with 1 byte
- 1 kilobyte = 1 KB = 1024 bytes about 10^3 bytes
  - A typical word processor document is about 100 KB
- 1 megabyte = 1 MB = 1024 KB - about 10^6 bytes
  - Digital photographs are usually anywhere from 1 to 5 MB
- 1 gigabyte = 1 GB = 1024 MB - about 10^9 bytes
  - a digital move file is about 1 GB

Capacity Lab Exercises

You don't need a calculator, just move the decimal points.

Q9 Express 256,000,000 bytes in KB

Q10 Express 256,000,000 bytes in MB

Q11 Express 256,000,000 bytes in GB

Capacity Lab Exercises

You don't need a calculator, just manipulate the exponent.

Q12 Express 3.0×10^6 bytes in KB

Q13 Express 3.0×10^6 bytes in MB

Q14 Express 3.0×10^6 bytes in GB

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What is throughput?

- Throughput is a measure of the amount of work completed per unit of time.
- Example:
  Willy Wonka's Chocolate Factory produces 600 everlasting gobstoppers every 4 hours.
  What is the throughput of gobstoppers per hour?
  600 gobstoppers / 4 hours =
  150 gobstoppers / hour
- Example:
  Throughput of 100 gallons of chocolate / minute
  How many gallons are processed in 10 hours?
  100 x 60 x 10 =
  60,000 gallons

Bus Bandwidth

- Bandwidth is the speed which information can be moved along a bus.
- Bandwidth = amount of information moved / unit time
- Bandwidth is measured in bytes, KB, or MB per second
- Examples
  - 100 KB per second
  - 2 GB per second
- What factors determine the peak or maximum bandwidth of a bus?
- Consider the tale of the fire-fighting gondoliers ...

Fire-fighting Gondoliers

- There is a fire in Venice
- The 16 fire fighters need to cross a canal to get to the fire.
- One gondola and its gondolier can ferry one fire fighter and his or her equipment in 10 minutes.
- How long will it take to move all 16 fire fighters?
  - 16 x 10 minutes = 160 minutes, ignoring the empty last return trip.
- What is the bandwidth?
  - 16 / 160 = 0.1 fire fighter per minute.

Fire-fighting Gondoliers: Speedup #1

- The gondolier realizes that more help is needed and so he calls 7 of his gondolier pals to bring their boats.
- Now 8 fire fighters can cross the canal at the same time.
- How long will it take to move all 16 fire fighters?
  - 2 x 10 minutes = 20 minutes
- What is the bandwidth?
  - 16/20 = 0.8 fire fighter per minute (8 times faster)

Fire-fighting Gondoliers: Speedup #2

- The mayor runs up screaming that his cat is stuck in the burning building and must be saved in 5 minutes.
- Is there any way to get a fire fighter across in time?
  - Speed up the transfer speed
- Suppose the 8 gondoliers had 8 motor boats that can ferry one fire fighter and his or her equipment in 4 minutes.
- How long will it take to move all 16 fire fighters?
  - 2 x 4 minutes = 8 minutes
- What is the bandwidth?
  - 16/8 = 2.0 fire fighters per minute

Bandwidth Summary

- Increase bandwidth by increasing the "width" of the bus or the number of lanes of the highway to move more information at the same time.
  - Typical PC buses transport information in chunks of 8, 16, 32, 64, 128, or 256 bytes.
- Increase the speed of travel on the bus("raise the speed limit")
  - Bus speed is rated in cycles per second
  - On cycle represents one transfer of information
    - (one gondola crossing)

Bandwidth Examples

- If a bus runs at 100 million clock cycles per second (100 MHz) and it delivers 8 bytes on each clock cycle, what is it's bandwidth?
  - 800 million bytes per second = 800 MB / second
- If a bus runs at 150 MHz and delivers 8 bytes per clock cycle, then what is its bandwidth?
  - 8 x 150 million = 1,200 MB / second = 1.2 GB / second

Bandwidth Lab Exercises

Q15 What is the bandwidth expressed in bytes/second of an 8-bit wide bus running at 10 MHz?

Q16 What is the bandwidth expressed in MB/second of a 32-bit wide bus running at 10 MHz?

Q17 What speed must an 8-bit wide bus have in order to equal the bandwidth of a 64-bit wide bus running at 800 MHz.

Peak versus Sustained Bandwidth

- The previous bandwidth calculations represent the theoretical or peak bandwidth assuming that a constant stream of data is always being pumped one-way through the bus
- In practice a bus services two way traffic
  - For example, send data from CPU to RAM or from RAM to CPU
- In Practice there is some overhead involved in preparing to send and in receiving a chunk of data
  - This overhead is called latency
- Consequently, in practice the actual or sustained bandwidth of a bus is less than its theoretical or peak bandwidth.

Latency and the Fire-fighting Gondoliers

- Our estimate of 10 minutes to transmit one fire fighter from one side of the canal to the other was greatly simplified
- Consider how that 10 minutes might be spent:
  - Load fire fighter and gear: 1.5 minutes
  - Cross the canalwith the fire fighter: 4 minutes
  - Unload fire fighter and gear: 1.5 minutes
  - Empty boat returns to the fire station: 3 minutes
- Only 4 of the 10 minutes is spent actually moving a fire fighter
  - Two-way trip represents reality of bus needing to handle two way transmissions data, only one of which goes at a time
  - loading/unloading represents latency (overhead)

Multiple Buses

- Having a USB digital camera, hard drive and video card sharing the same bus would be like having pedestrians, bicyclists, and automobiles sharing the same road
- Modern PC motherboards include more than one bus to accommodate traffic flowing at vastly different speeds
- Use a separate switch and bus line for each different type of traffic
  - USB bus (slowest)
  - Expansion device bus for hard drive, CS, DVD, network card
  - Video bus (PCI-E 16x)
  - Frontside bus - connects CPU and RAM

Hand in Your Lab

Be sure to format your answers so they are easy to read.
Save the document as Lec03.txt and submit it on Brightspace.